Algorithms and Mathematical Concepts for Computing

2nd September 2024

Examination Paper

Answer ALL questions.

Clearly cross out surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Answer ALL questions

Marks

Question 1

Consider the following scenario:

You are tasked with developing a software application for a small business to manage their inventory. The application needs to track product quantities, update inventory levels and generate reports.

a) State in which stage of the software development process are detailed technical specifications and architectural plans created based on the gathered requirements and defined scope? **1**

b) State which stage of the software development process involves translating the design into a functioning system by writing code and integrating various components. **1**

c) Explain the concepts of abstraction and decomposition in problem-solving and how they can be applied when designing software solutions. You must include an explanation of the purpose and benefits of both abstraction and decomposition. Provide one real-life example for each concept (abstraction and decomposition). **8**

Total 10 Marks

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Question 2

Consider the following content related to numbers and number systems:

a) Which of the following is **NOT** a type of number? **1**

i) Integer
ii) String
iii) Natural number
iv) Irrational number

b) Decimal to _____ conversion involves converting a number from base 10 to base 8. **1**

c) Describe the concept of prime factorisation and explain its significance in identifying prime numbers. **2**

d) Perform the following conversions:

i) Convert the binary number 1011 to octal number. **2**

ii) Convert the octal number 45 to binary number. **2**

iii) Convert the decimal number 13 to Hexadecimal number. **2**

Total 10 Marks

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Question 3

Answer the following questions related to logic and Boolean algebra:

a) Which of the following operators represents the logical AND operation? **1**

i) ^
ii) &
iii) |
iv) !

b) Complete the truth table for the Boolean expression: **6**

$$(P \vee Q) \wedge (\neg P)$$

P	**Q**	$\neg P$	$P \vee Q$	$(P \vee Q) \wedge (\neg P)$
T	T			
T	F			
F	T			
F	F			

c) Explain the concept of conditional statements in Boolean algebra and provide an example. **3**

Total 10 Marks

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Question 4

Answer the following questions related to Predicate Logic, Set Theory and Venn Diagrams:

a) Which of the following symbols represents the existential quantifier in predicate logic? **1**

- ∀
- ∃
- ∧
- ∨

b) Explain the concept of set intersection and provide an example **2**

c) Calculate the complement of the following set: **1**

A = {1, 2, 3, 4} in the universal set {1, 2, 3, 4, 5, 6}.

d) What is the Cartesian product of two sets and how is it represented? Calculate the Cartesian product of the following sets: A = {4, 5} B = {x, y, z} **4**

e) Calculate the union of the following sets: **2**

A = {a, b, c}
B = {c, d, e}

Total 10 Marks

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Question 5

Answer the following questions related to statistics:

a) Discuss TWO (2) of the measures of central tendency and provide examples of situations where they are appropriate or inappropriate to use. **2**

b) In statistics, _____ is a measure of how spread out the values in a data set are around the mean. **1**

c) Calculate the median of the following data set: 25, 20, 30, 10, 15, 35. Show your working. **2**

d) Explain the concept of quartiles in statistics. **2**

e) Calculate the mean of the following data set: 15, 27, 18, 21,12, 24, Show your working. **3**

Total 10 Marks

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Question 6

Answer the following questions related to Data Structures and Arrays:

a) Which of the following is **NOT** a characteristic of a data structure? **1**

i) Used to organise and store data in a computer system.
ii) A programming language used for data manipulation.
iii) A tool for analysing data patterns.
iv) A graphical representation of data relationships.

b) In data structures, a _____ is a linear collection of elements, where each element points to the next one in the sequence. **1**

c) Define the data type "array" and explain its role in storing data. **3**

d) Write pseudocode to find the largest element in a one-dimensional array. Assume the existence of a function called ***length(arr)*** which returns the number of elements in the array ***arr***. **5**

Total 10 Marks

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Question 7

Answer the following questions related to Data Structures:

a) Which data structure follows the First-In-First-Out (FIFO) principle? **1**

i) Stack
ii) Queue
iii) Linked List
iv) Array

b) In a stack, elements are removed from the _____ and added to the _____. **2**

c) Consider a queue initially containing the elements 10, 20, 30. Assume 10 is at the front of the queue and 30 is at the back. Perform the following operations one after the other: **3**

I. Dequeue one element.
II. Enqueue the number 40.
III. Dequeue two elements.

Show the final contents of the queue after these operations.

Show your working and draw out the queue after each operation.

d) Provide **ONE (1)** real-world example where each of a stack and a queue data structure would be used. **4**

Total 10 Marks

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Question 8

Answer the following questions related to algorithm efficiency and searching techniques:

a) Which of the following best describes **Big O** notation in the context of algorithm analysis? **1**

i) The best-case scenario of an algorithm's time complexity.
ii) The average-case performance of an algorithm.
iii) The upper bound on the growth rate of the algorithm's runtime.
iv) The precise execution time of an algorithm for a given input size.

b) Binary search algorithms have a time complexity of _____. **1**

c) Write pseudocode for a function for a **linear search** algorithm that searches for a target value in a 1-dimensional array. The function takes the target value and array as parameters. It returns the index of the target value or -1 if the target is not found. **6**

You should assume the existence of a function called ***length(arr)*** which returns the number of elements in the array ***arr***.

d) Explain why binary search is more efficient than linear search for sorted arrays. **2**

Total 10 Marks

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Question 9

Answer the following questions related to sorting algorithms and recursion:

a) Which sorting algorithm works by repeatedly selecting the smallest (or largest) element from the unsorted portion of the array and placing it in its correct position in the sorted portion? **1**

i) Bubble Sort
ii) Quick Sort
iii) Merge Sort
iv) Insertion Sort

b) State what is meant by recursion in programming. **1**

c) Fill in the blanks in the pseudocode for the ***quickSort*** function implementing the quick sort algorithm. You can assume the existence of the following functions. **6**

- ***length(arr)***: Returns the number of elements in the array ***arr***.
- ***partition (arr, low, high)***: Partitions the array ***arr*** based on a pivot element and returns the index of the pivot.

```
Function quickSort(arr, low, high):
    if _____:
        # Partition the array
        pivotIndex = _____(arr, low, high)

        # Recursively sort the left and right partitions
        quickSort(arr, low, _____)
        quickSort(arr, _____, high)
    End function
```

d) Explain how recursion applies to the process of calculating factorial numbers. **2**

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Question 10

Answer the following questions related to tree data structures:

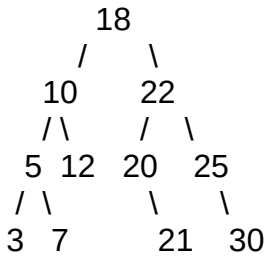
a) Which of the following best describes a binary tree data structure? **1**

i) A hierarchical structure composed of nodes connected by edges.
ii) A structure where elements are arranged in a linear sequence.
iii) A data structure that stores elements in a hash table.
iv) A structure where elements are stored in a stack.

b) Fill in the missing part of the following statement: **1**
In a binary tree, each node has a maximum of _____ children.

c) Construct a binary search tree from the following sequence of numbers: 8, 4, 12, 2, 6, 10, 14. Ensure that 8 is the root of the tree. **4**

d) Using the provided binary search tree (BST) below, demonstrate the steps involved in searching for the value 22. Begin the search at the root node and describe each comparison and movement until the value is found or determined to be absent. **4**



Total 10 Marks

End of paper

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